Dvodimenzijske kontinuirane slučajne varijable – zavisne

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1 Funkcija gustoće (vjerojatnosti)

$$f(x,y) = x + y, \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1 \qquad (0 \text{ inače})$$

Provjera normiranosti:

$$\int_0^1 \int_0^1 (x+y)\,dx\,dy = \int_0^1 \left(\frac{1}{2} + y\right) dy = \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark$$

2 Marginalne funkcije gustoće

$$f(x) = \int_c^d f(x,y)\,dy$$

$$f(x) = \int_0^1 (x+y)\,dy \tag{1}$$

$$f(x) = x + \int_0^1 y\,dy \tag{2}$$

$$f(x) = x + \frac{y^2}{2}\Big|_0^1 \tag{3}$$

$$f(x) = x + \frac{1}{2}, \qquad 0 \leq x \leq 1 \tag{4}$$

$$f(y) = \int_0^1 (x+y)\,dx \tag{5}$$

$$f(y) = y + \frac{1}{2}, \qquad 0 \leq y \leq 1 \tag{6}$$

3 Uvjetne funkcije gustoće

$$f(x \mid y) = \frac{f(x,y)}{f(y)}$$

$$f(x \mid y) = \frac{x+y}{y+\frac{1}{2}} \tag{7}$$

$$f(x \mid y) \neq f(x) = x + \frac{1}{2} \longrightarrow \text{zavisne varijable} \tag{8}$$

$$f(y \mid x) = \frac{f(x,y)}{f(x)} = \frac{x+y}{x+\frac{1}{2}} \tag{9}$$

$$f(y \mid x) \neq f(y) = y + \frac{1}{2} \longrightarrow \text{zavisne varijable} \tag{10}$$

$$f(x,y) = x + y \neq f(x)f(y) = \left(x + \frac{1}{2}\right)\left(y + \frac{1}{2}\right) \longrightarrow \text{zavisne varijable}$$

4 Matematička očekivanja i varijance

4.1 Matematičko očekivanje

$$E(X) = \int_a^b x f(x)\, dx$$

$$E(X) = \int_0^1 x\left(x + \frac{1}{2}\right) dx \tag{11}$$

$$E(X) = \int_0^1 \left(x^2 + \frac{x}{2}\right) dx \tag{12}$$

$$E(X) = \left(\frac{x^3}{3} + \frac{x^2}{4}\right)\Big|_0^1 \tag{13}$$

$$E(X) = \frac{1}{3} + \frac{1}{4} \tag{14}$$

$$E(X) = \frac{7}{12} \tag{15}$$

$$E(Y) = \int_0^1 y\left(y + \frac{1}{2}\right) dy \tag{16}$$

$$E(Y) = \frac{7}{12} \tag{17}$$

4.2 Varijanca

$$V(X) = \int_a^b \big(x - E(X)\big)^2 f(x)\, dx = \int_a^b x^2 f(x)\, dx - E(X)^2$$

$$V(X) = \int_0^1 x^2\left(x + \frac{1}{2}\right) dx - \left(\frac{7}{12}\right)^2 \tag{18}$$

$$V(X) = \int_0^1 \left(x^3 + \frac{x^2}{2}\right) dx - \frac{49}{144} \tag{19}$$

$$V(X) = \left(\frac{x^4}{4} + \frac{x^3}{6}\right)\Big|_0^1 - \frac{49}{144} \tag{20}$$

$$V(X) = \frac{1}{4} + \frac{1}{6} - \frac{49}{144} \tag{21}$$

$$V(X) = \frac{11}{144} \tag{22}$$

$$V(Y) = \int_0^1 y^2 \left(y + \frac{1}{2}\right) dy - \left(\frac{7}{12}\right)^2 \tag{23}$$

$$V(Y) = \frac{11}{144} \tag{24}$$

5 Uvjetno očekivanje, kovarijanca i koeficijent korelacije

5.1 Uvjetno očekivanje

$$E(X \mid y) = \int_a^b x f(x \mid y)\, dx$$

$$E(X \mid y) = \int_0^1 x \, \frac{x+y}{y+\frac{1}{2}}\, dx \tag{25}$$

$$E(X \mid y) = \frac{1}{y+\frac{1}{2}} \int_0^1 \left(x^2 + xy\right) dx \tag{26}$$

$$E(X \mid y) = \frac{1}{y+\frac{1}{2}} \left(\frac{x^3}{3} + \frac{yx^2}{2}\right)\Bigg|_0^1 \tag{27}$$

$$E(X \mid y) = \frac{1}{y+\frac{1}{2}} \left(\frac{1}{3} + \frac{y}{2}\right) \tag{28}$$

$$E(X \mid y) = \frac{\frac{1}{3} + \frac{y}{2}}{y+\frac{1}{2}} \tag{29}$$

$$E(X \mid y) = \frac{3y+2}{6y+3} \longrightarrow \text{zavisne varijable, regresija nije linearna} \tag{30}$$

Kontrola: za $y = 0$ je $f(x \mid 0) = \dfrac{x}{1/2} = 2x$, pa je $E(X \mid 0) = \dfrac{2}{3}$, što formula i daje: $\dfrac{3 \cdot 0 + 2}{6 \cdot 0 + 3} = \dfrac{2}{3}$. ✓

$$E(Y \mid x) = \int_0^1 y \, \frac{x+y}{x+\frac{1}{2}}\, dy \tag{31}$$

$$E(Y \mid x) = \frac{3x+2}{6x+3} \longrightarrow \text{zavisne varijable, regresija nije linearna} \tag{32}$$

5.2 Kovarijanca

$$\mathrm{Cov}(X,Y) = \int_a^b \int_c^d \big(x - E(X)\big)\big(y - E(Y)\big) f(x,y)\, dx\, dy$$

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$$\mathrm{Cov}(X,Y) = \int_a^b \int_c^d xy\, f(x,y)\, dx\, dy - E(X)E(Y)$$

$$\mathrm{Cov}(X,Y) = \int_0^1 \int_0^1 xy\,(x+y)\,dx\,dy - \frac{7}{12}\cdot\frac{7}{12} \tag{33}$$

$$= \int_0^1 \int_0^1 \left(x^2 y + xy^2\right) dx\,dy - \frac{49}{144} \tag{34}$$

$$= \left(\frac{1}{3}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{3}\right) - \frac{49}{144} \tag{35}$$

$$= \frac{1}{3} - \frac{49}{144} = \frac{48}{144} - \frac{49}{144} \tag{36}$$

$$\mathrm{Cov}(X,Y) = -\frac{1}{144} \tag{37}$$

5.3 Koeficijent korelacije

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma(X)\,\sigma(Y)}$$

$$\sigma(X) = \sqrt{V(X)} = \sqrt{\frac{11}{144}} = \frac{\sqrt{11}}{12} \tag{38}$$

$$\sigma(Y) = \sqrt{V(Y)} = \frac{\sqrt{11}}{12} \tag{39}$$

$$\rho = \frac{-\frac{1}{144}}{\frac{\sqrt{11}}{12}\cdot\frac{\sqrt{11}}{12}} = \frac{-\frac{1}{144}}{\frac{11}{144}} \tag{40}$$

$$\rho = -\frac{1}{11} \tag{41}$$

Napomena: Koeficijent korelacije ρ uvijek je dobro definiran i mjeri *linearnu* povezanost varijabli. Budući da ovdje funkcije regresije nisu linearne, ρ ne opisuje potpunu (nelinearnu) zavisnost, nego samo njezinu linearnu komponentu, pa ga treba interpretirati s oprezom.